

Model Development Document

Stage-1 Regulatory Gate — CMPL-REG-24

EDA · 80/10/10 split · LogReg vs XGBoost vs LightGBM vs BERT
validation-optimized cut-off · OOV analysis · sensitivity analysis

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Data: data/complaints/cfpb_complaints.csv (4,000 rows)

Split: 5% scoring holdout, then 80/10/10 stratified, seed 42

Champion: lightgbm @ cut-off 0.992

Repo: reg-agents - model CMPL-REG-24 stage 1

Artifacts: docs/model_development/artifacts/

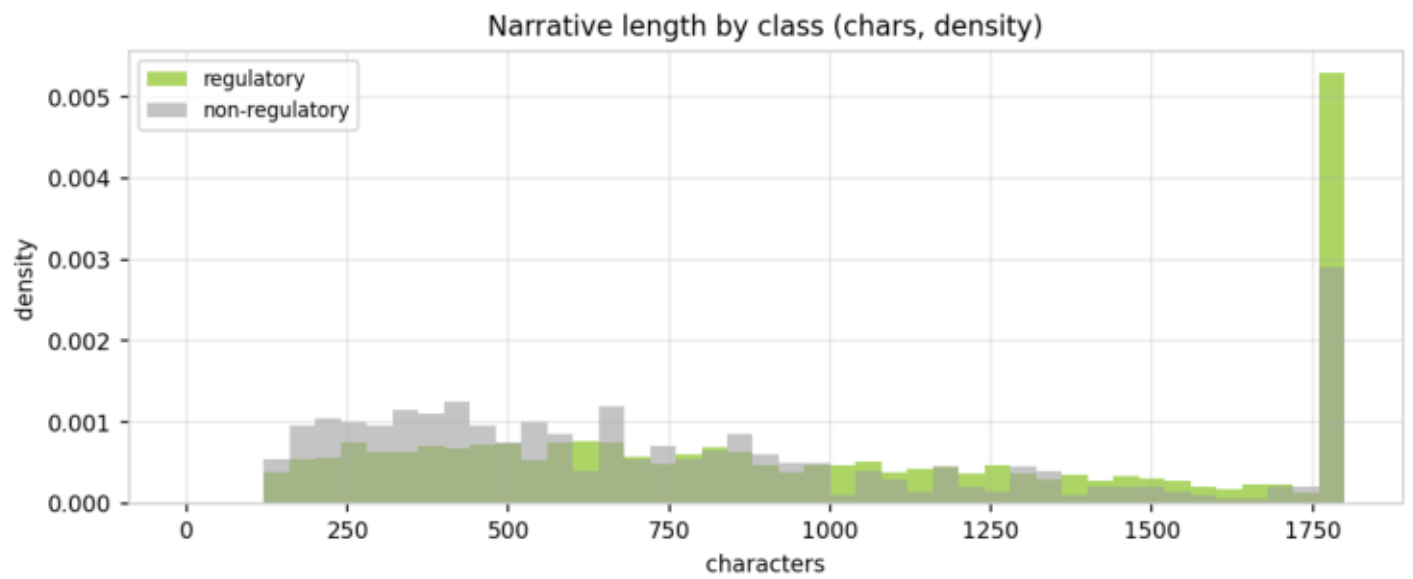
Abstract

This document records the development of the stage-1 regulatory gate for CMPL-REG-24 on 4,000 curated CFPB complaint narratives (87.5% regulatory, 12.5% non-regulatory minority). Four minority-balanced classifiers — logistic regression, XGBoost, LightGBM, and a fine-tuned DistilBERT — were compared under a stratified 80/10/10 train/validation/test protocol (applied after reserving a 5% scoring holdout for the batch-ingestion layer), with selection on validation minority-class PR-AUC, a decision cut-off optimized on the validation fold (0.992 for the champion, in place of the default 0.5), and one-shot test reporting. The champion, lightgbm, achieved validation minority PR-AUC 0.6978 and test minority PR-AUC 0.6908 (ROC-AUC 0.9367). Out-of-vocabulary and sensitivity analyses (threshold sweep, input perturbations, class-weight ablation, split-seed stability) bound the model's robustness. Reference labels are weak supervision from the CFPB taxonomy, and with only ~47 minority cases per held-out fold, minority metrics carry material split variance; both caveats condition the deployment recommendation.

2 · Data — class balance (Table 1) & properties (Table 2)

class / property	n / value	share
regulatory (1)	3,500	87.5%
non-regulatory (0)	500	12.5%
chars — reg / non-reg	911 / 656 (median)	
words — reg / non-reg	168 / 122 (median)	
'XXXX' PII masks per narrative	8.7 (mean)	

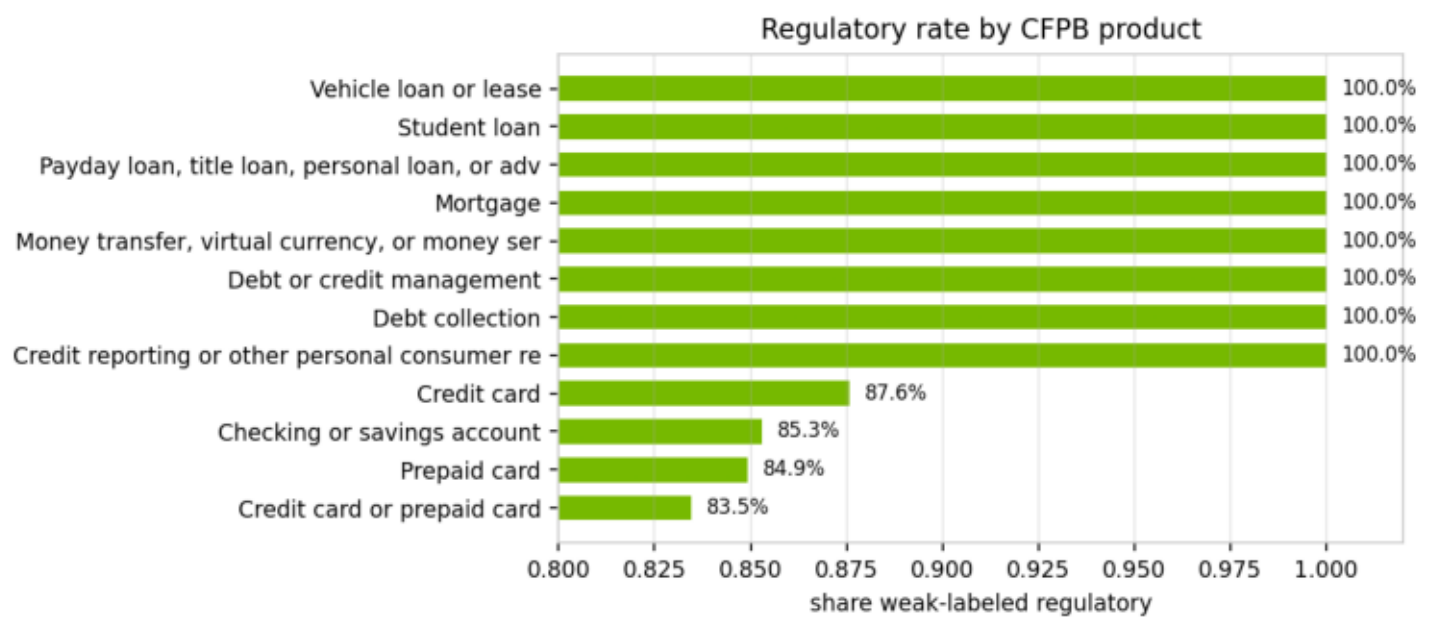
3 · EDA — narrative length by class (Fig. 1)



3 · EDA — regulatory rate by product (Table 3)

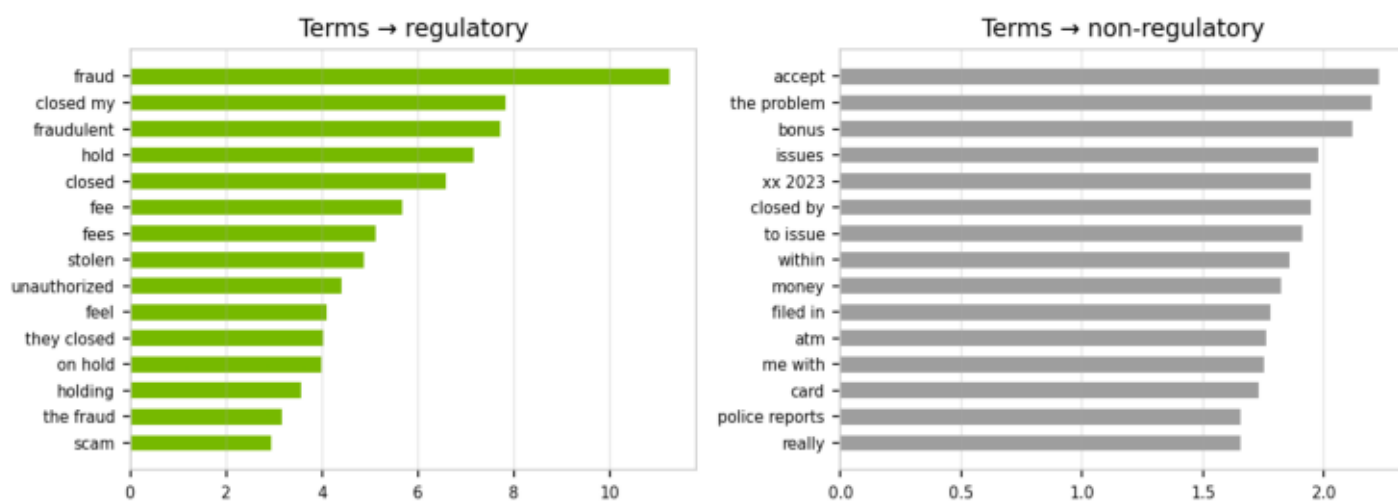
product	n	regulatory rate
Credit card or prepaid card	508	83.5%
Prepaid card	126	84.9%
Checking or savings account	2,068	85.3%
Credit card	748	87.6%
Credit reporting or other personal c	449	100.0%
Debt collection	55	100.0%
Debt or credit management	3	100.0%
Money transfer, virtual currency, or	11	100.0%
Mortgage	13	100.0%
Payday loan, title loan, personal loa	5	100.0%
Student loan	4	100.0%
Vehicle loan or lease	10	100.0%

3 · EDA — products (Fig. 2)



3 · EDA — discriminative terms (Fig. 3)

Most discriminative TF-IDF terms (balanced logistic coefficients)



4 · Split — stratified 80/10/10, seed 42 (Table 5)

split	rows	regulatory	non-regulatory	reg rate
train (80%)	3,040	2,660	380	87.50%
validation (10%)	380	333	47	87.63%
test (10%)	380	332	48	87.37%
scoring holdout (5% c	200	175	25	87.50%

5 · Candidates and hyperparameters (Table 6)

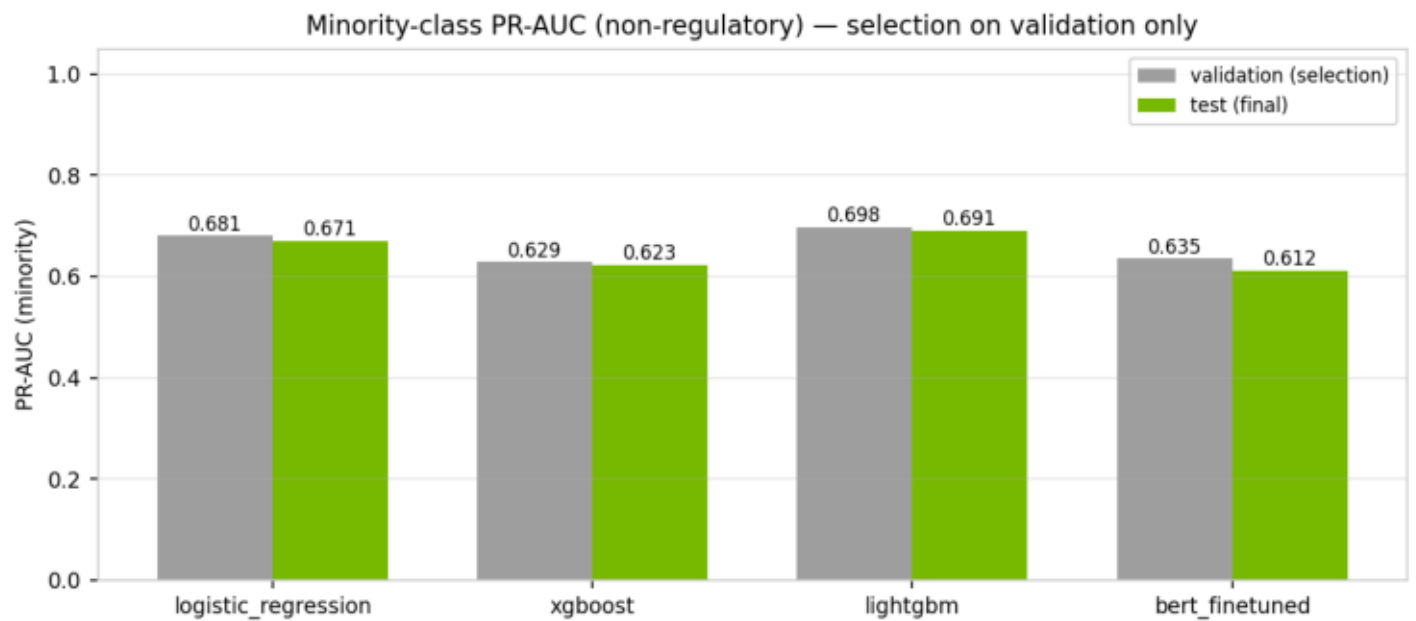
model	representation	hyperparameters	fit (s)	
logistic_regression	TF-IDF 30k, 1-2 gram, subv	L1, C=4.0 (validation-tune	0.06	train/test ROC g
xgboost	same TF-IDF features	300 trees, depth 6, lr 0.1, l	2.68	=0.1
lightgbm	same TF-IDF features	400 trees, 63 leaves, lr 0.0	10.05	
bert_finetuned	raw text → WordPiece subv	distilbert-base-uncased, m	103.7	W lr 3e-5, 2 epo

6 · Leaderboard (Table 7)

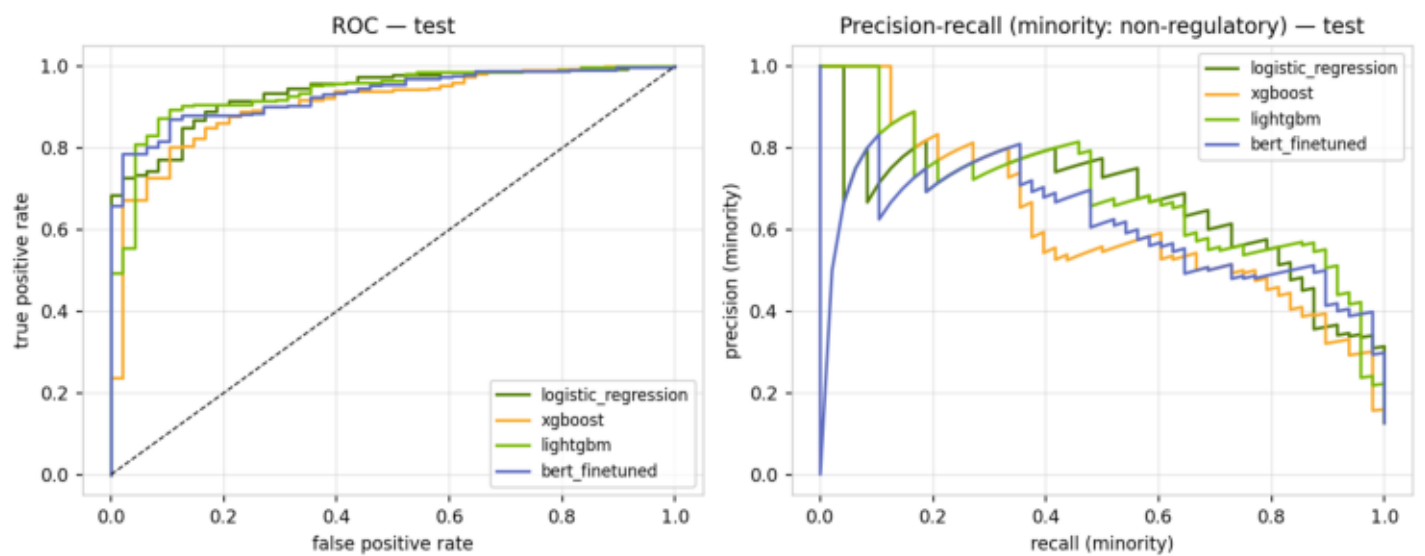
Champion: lightgbm - selected on validation minority PR-AUC; other columns are one-shot test metrics.

model	PR-AUC min (val)	cut-off (val)	PR-AUC min (test)	ROC-AUC (test)	F1 min (test)	bal-acc (test)	Brier (test)	fit (s)
logistic_reg	0.6814	0.491	0.6707	0.9362	0.6422	0.8254	0.0741	0.06
xgboost	0.6294	0.545	0.6234	0.9073	0.5979	0.772	0.0746	2.68
lightgbm	0.6978	0.992	0.6908	0.9367	0.6261	0.8283	0.0745	10.05
bert_finetune	0.6355	0.478	0.6116	0.9317	0.5854	0.8163	0.088	103.7

6 · Minority PR-AUC by model (Fig. 4)



6 · ROC and minority PR curves, test (Fig. 5)



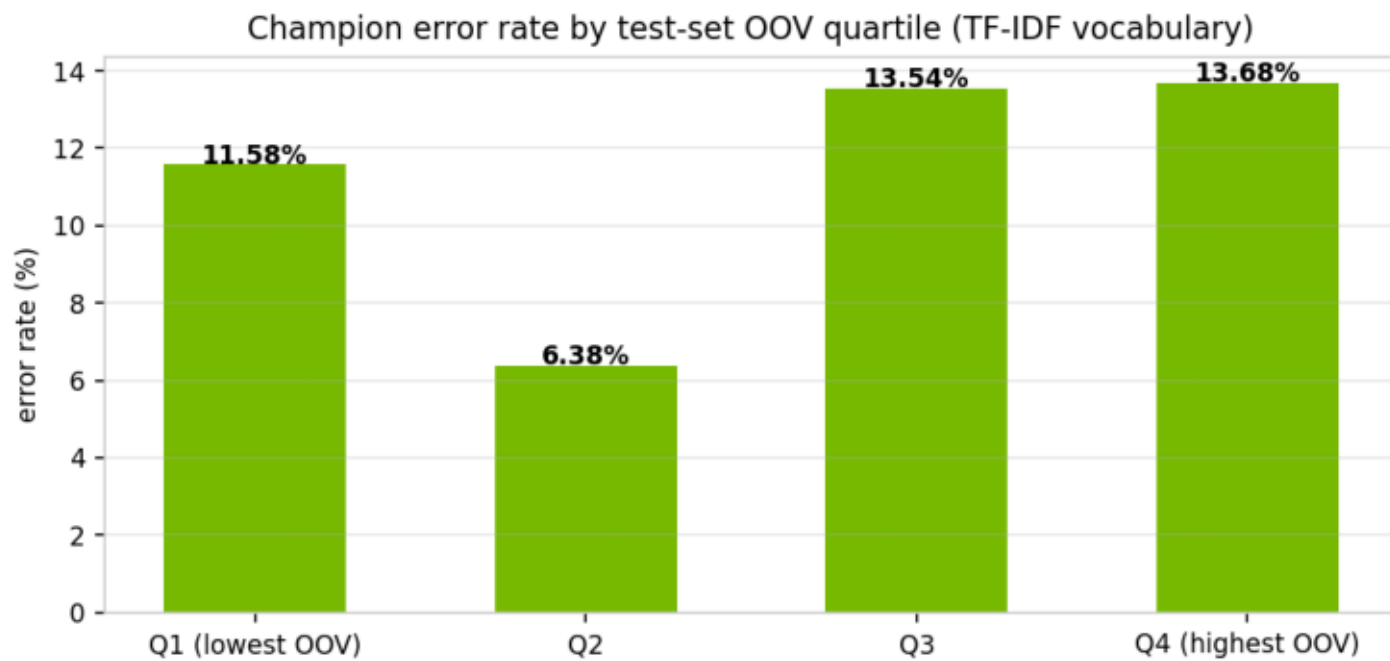
6 · Champion confusion matrix, test @ cut-off 0.992 (Table 8)

actual \ predicted	non-regulatory	regulatory
actual non-regulatory	36	12
actual regulatory	31	301

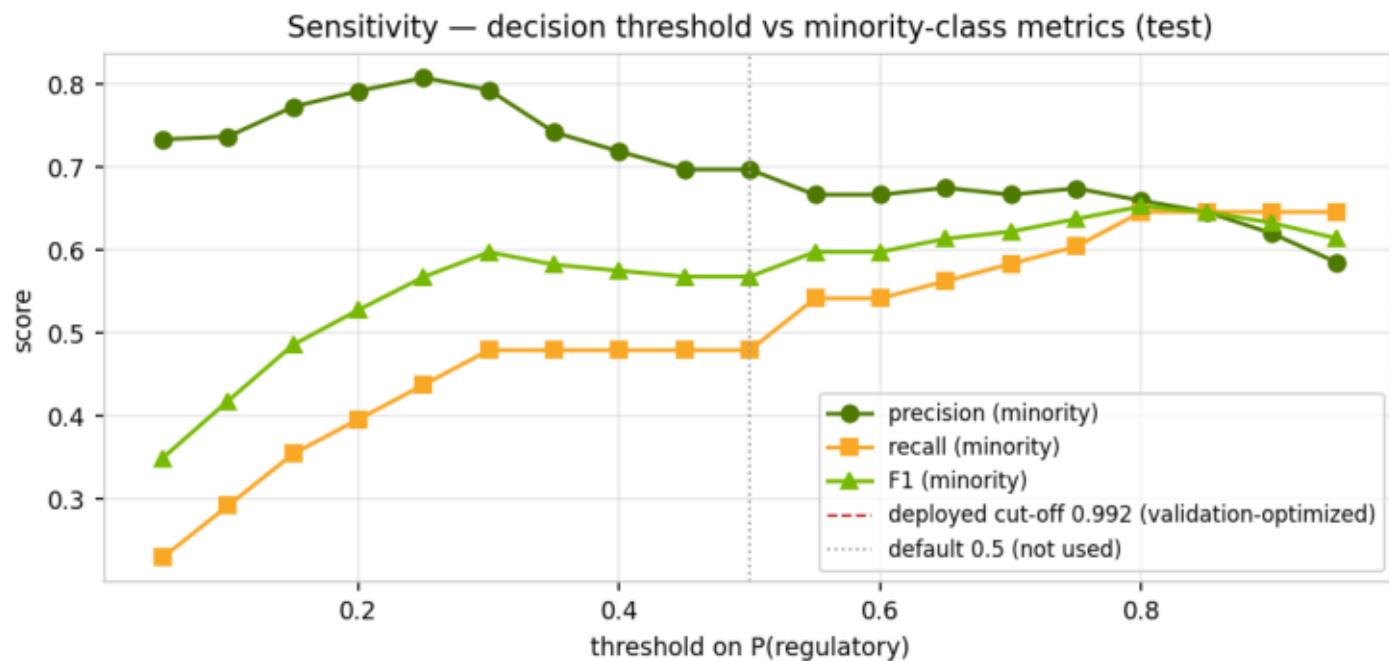
7 · OOV exposure (Tables 9-10)

OOV quartile / representative	rate	n	error rate
TF-IDF vocabulary (30k, train)	1.9% of test tokens out-of-		
DistilBERT subword tokenization	0.000% of test tokens mapped		
Q1 (lowest OOV)	0.0%	95	11.58%
Q2	0.8%	94	6.38%
Q3	1.8%	96	13.54%
Q4 (highest OOV)	4.3%	95	13.68%

7 · OOV analysis (Fig. 6)



8 · Threshold sensitivity (Fig. 7)



8 · Perturbation sensitivity (Table 11)

perturbation	PR-AUC (minority)	Δ
baseline (unperturbed)	0.6908	—
remove CFPB 'XXXX' PII masks	0.6925	+0.0016
lowercase + strip punctuation	0.6908	+0.0000
keep only first 50% of narrative	0.3565	-0.3344
randomly drop 10% of tokens	0.6622	-0.0286

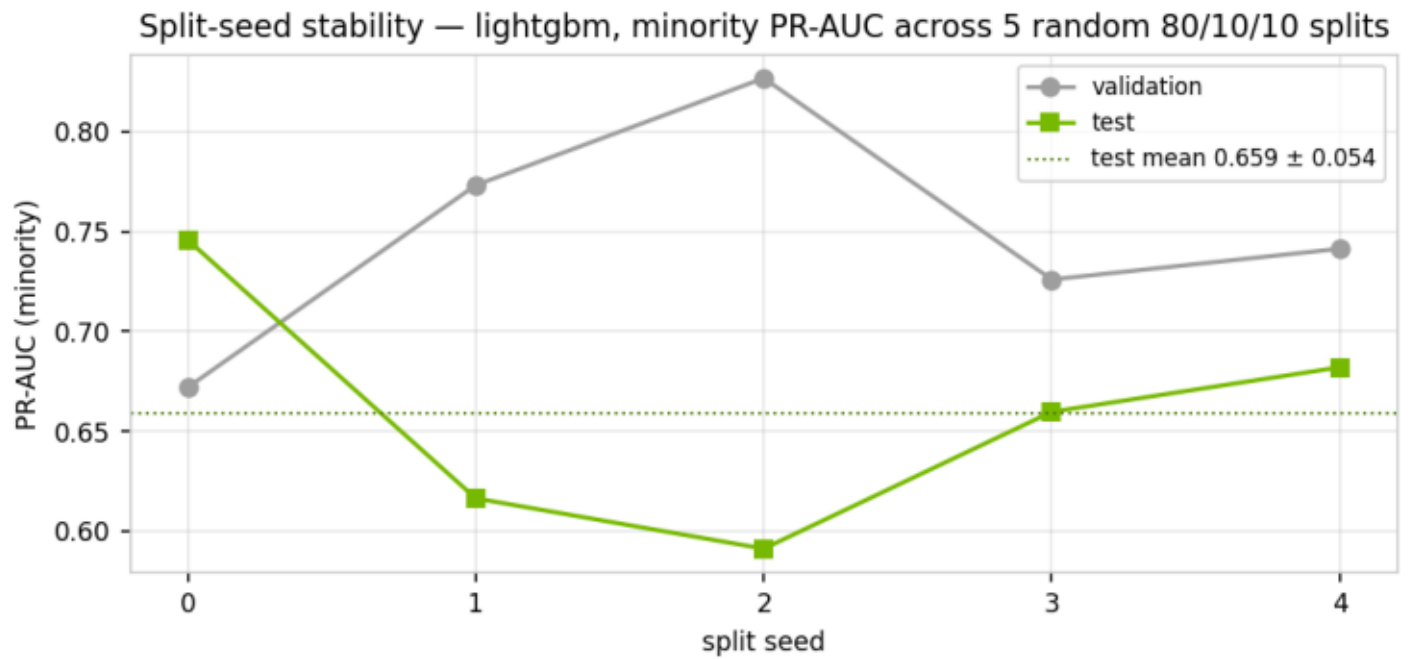
8 · Class-weight ablation (Table 12)

weighting	cut-off	PR-AUC min (val)	PR-AUC min (test)	F1 min (test)	bal-acc (test)
balanced (deploy)	0.992	0.6978	0.6908	0.6261	0.8283
unweighted	0.949	0.6991	0.6933	0.6122	0.7824

8 · Split-seed stability (Table 13)

seed	PR-AUC min (val)	PR-AUC min (test)
seed 0	0.6720	0.7458
seed 1	0.7733	0.6163
seed 2	0.8269	0.5911
seed 3	0.7260	0.6597
seed 4	0.7414	0.6820
mean $\pm$ std	0.7479 $\pm$ 0.0513	0.6590 $\pm$ 0.0538

8 · Seed stability (Fig. 8)



9 · Leakage audit — split contamination (Table 14)

Curation dedups exact hashes, 200-char prefixes and TF-IDF-cosine near-copies before any split.

check (test vs train)	count	expected	status
exact duplicate (normalize	0	0	pass
near-duplicate (200-char p	0	0	pass
near-duplicate (TF-IDF cosi	2	$\leq 1\%$ of test	pass

9 · Leakage audit — label provenance slices (Table 15)

75% of labels are regex-derived from the narrative (weak-supervision target leakage); the metadata-labeled slice is the leak

test slice	n	reg-rate	ROC-AUC	PR-AUC (reg)	F1 (min)	bal acc
full test set (he	380	0.87	0.937	0.990	0.626	0.828
metadata-label	81	0.41	0.907	0.895	0.818	0.814
narrative-regex	299	1.00	— (single class	—	—	—

10 · Discussion and model selection decision

The bake-off selected lightgbm on validation minority PR-AUC. The spread between validation and test metrics, and the seed-stability band (0.7479 ± 0.0513 validation, 0.6590 ± 0.0538 test), shows that with only ~47 minority cases per held-out fold, single-split point estimates should not be over-read: the candidates are statistically close, and simplicity, latency, and interpretability are legitimate tie-breakers. The decision cut-off was optimized on the validation fold by maximizing minority F1 — the champion deploys at 0.992 rather than the default 0.5, which sits at a poorer point on the minority precision-recall trade-off (Figure 7). Perturbation analysis shows the model is robust to PII-mask removal and case/punctuation noise but degrades when narratives are truncated, so upstream text ingestion must preserve full narratives. OOV exposure is modest (1.9% of test tokens) and does not concentrate errors in the highest-OOV quartile. The class-weight ablation confirms minority balancing materially improves minority F1 at each variant's own optimized cut-off. The fine-tuned DistilBERT does not clear the bar to justify GPU serving cost for this gate today; it remains the upgrade path if complaint language drifts (subword tokenization eliminates OOV) or if golden-set labels reveal weak-label ceiling effects. Recommendation: lightgbm with the balanced weighting at the 0.992 cut-off is the research champion and promotion candidate; the deployed production gate keeps the champion and cut-off of the committed governance run (docs/complaint_model/metrics.json), with promotion via the standard change-management path. Revisit selection when the golden set lands or PSI drift alerts fire.

12 · Artifact manifest (Table 16)

artifact	size	description
tfidf_vectorizer.joblib	328 KB	TF-IDF featurizer (30k, 1-2 gram, sublinear, min_df
model_logistic_regression.joblib	6 KB	fitted candidate (challenger)
model_xgboost.joblib	125 KB	fitted candidate (challenger)
model_lightgbm.joblib	1,322 KB	fitted candidate (CHAMPION)
split_indices.json	41 KB	exact train/validation/test membership (reproducib
environment.txt	0 KB	package versions used for this run
(bert_finetuned weights)	—	fine-tuned distilbert-base-uncased (epochs=2, max

12 · Environment (Table 17)

package	version
python	3.11.15
numpy	1.26.4
pandas	2.2.3
scikit-learn	1.5.2
xgboost	2.1.1
lightgbm	4.7.0
torch	2.13.0
transformers	5.14.1
matplotlib	3.9.2

Sign-off

Prepared under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Independent review: submitted to second-line validation with this document, the data profile, and the artifact set under `docs/model\_development/artifacts/`.